Optimizing Sales, Inventory and Margin

For Eesbee Garments

Final report for the BDM capstone Project

Submitted by

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1. Executive summary

Eesbee is a B2C garment store located in Alkapuri, Hyderabad that sells a variety of branded items. A comprehensive review was conducted to evaluate margin performance, inventory flow, and sales efficiency across the business. Data collected from the POS system was cleaned, normalized and enhanced for accurate analysis. An analyzer engine was built in Python using statistical and visualization frameworks.

Multiple analytical methods like segmented contribution analysis and SKU velocity tracking were used to evaluate margin and sales. Inventory flow diagnostics were performed to identify discrepancies, negative stock positions, and unsold purchase returns. Unit price distribution and discount allocation were examined to assess pricing discipline, while a sales trend calendar was developed to highlight seasonal fluctuations and promotional impacts.

The analysis revealed that margin efficiency varied significantly across segments, with some contributing disproportionately to the overall profitability. Enhanced performer analysis showed that certain SKUs delivered consistent returns, while others exhibited varying returns due to discount pressure or stock mismanagement. Inventory diagnostics highlighted issues such as dead stock accumulation, stock discrepancies, and unsold returns, all of which constrained working capital and reduced margin realization. Discount distribution was found to be uneven, with excessive markdowns eroding profitability in specific months.

This final submission ties these findings into a structured improvement plan with recommendations and action items. By aligning margin efficiency, inventory accuracy, and sales discipline, profitability can be enhanced and operational risks can be reduced. The recommended actions provide a roadmap for sustainable performance improvement overtime for Eesbee across areas like inventory health, sales, margin and profitability.

2. Analysis Process, Methods and Tools

The following section describes the process flow, methods used and tools developed to achieve the goals stated in the *Proposal* and *Mid-Term* submissions.

2.1. Process Flow

The core work for the project is divided into three parts: Pre-processing, processing and the Conclusion stages, as shown in the following Figure **FSF-01**. Details of each stage and their activities were described in Mid-Term submission and have not changed; hence they are not repeated here for succinctness.

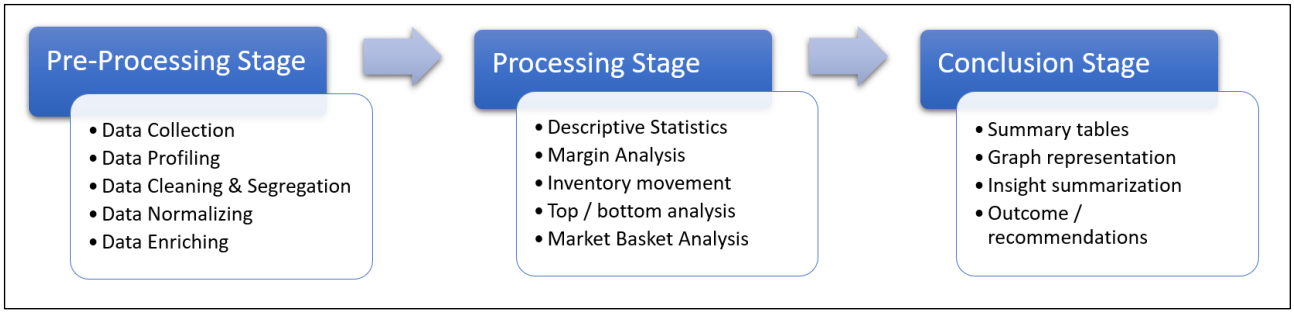


Figure **FSF-01**: Process Stages and their breakdown

2.2. Analytical Methods

This project adopts a modular, diagnostic-first approach to retail analytics, with each method aligned to a specific business objective. The emphasis is on being interpretable, reproducible, and relevant to the business with transparent logic and structured outputs.

Exploratory Data Analysis: ***EDA*** was conducted during initial profiling to assess data structure, volume, and integrity. Summary statistics and distribution plots were used to identify outliers, pricing anomalies, and return behavior, informing the design of downstream diagnostics.

Margin Diagnostics: Margin diagnostic dataset was computed using aggregated SKU on opening stock, purchase, purchase return, sales, customer returns and closing stock. This aggregated data was combined on SKU to create a unified stock movement view. Additional diagnostic information like net quantity, net value, unit cost, unit selling price were also computed for further analysis.

Realized Margin: Realized margin is the is the actual money realized by the business from sales before any expenses. It is computed by aligning transactional sales data with corresponding cost of goods sold. Customer returns were netted against sales to ensure accuracy in the realized outcome.

Unrealized Margin: Unrealized margin is derived from closing stock valuations, representing the potential margin embedded in the unsold inventory. It is a potential sale value, as the margin could reduce based on the discount given at the sale. This measure provided visibility into margin opportunities that remain locked in inventory.

Margin Gain: Margin gain was measured by isolating SKUs where potential margin at closing stock exceeded potential margin at opening stock. The positive variance was computed to capture incremental profitability generated through favorable stock movements, improved pricing, or better inventory mix.

Margin Erosion: Margin erosion was identified by comparing potential margin at opening stock with potential margin at closing stock. SKUs where closing margin fell below opening margin were flagged, and the negative variance was quantified. This method highlights the impact of slow-moving, discounted, or devalued stock on overall profitability.

***Segmented Contribution & Margin Efficiency*:** A segmented contribution and margin efficiency analysis was performed using percentile-ranked SKU groups based on total sales value. Normalized contributions for both sales (>0) and realized margin (>0) were visualized through dual-axis bar charts, while margin efficiency was overlaid as a line plot to highlight profitability per unit sale. The segmentation included top-performing, middle-core, and bottom-tier SKUs to enable comparative diagnostics. Axis labels and efficiency stabilizers were applied to ensure interpretability and suppress misleading ratios from low-volume segments.

Top performers: Top performers for Realized Gains, Sale Quantity and Sale Value are considered for *Leaderboards* as they help in driving topline performance. When measured by *sales quantity*, fast-moving items surfaced as volume leaders, highlighting their role in sustaining demand momentum. Analysis of *realized margin* revealed a distinct set of SKUs that delivered superior profitability, often diverging from pure *Sale value* leaders, thereby emphasizing the need to balance volume with margin contribution.

Inventory Flow Diagnostics: The inventory data was reviewed for the period from April 1st 2023 to September 30th 2025. The opening balance was taken as the starting point, and each movement of stock was traced through purchases, purchase returns, verification adjustments, sales, and discrepancies until the closing balance was reached. The values were checked to ensure that the additions and reductions reconciled correctly with the reported closing stock. Ratios such as inventory turnover and purchase-to-sales alignment were calculated so that the pace of inventory movement could be compared with common retail benchmarks. A waterfall chart was prepared to show how each activity contributed to the overall change in stock. This approach was used so that the flow of inventory could be understood not only in terms of raw numbers but also in relation to industry standards, allowing risks and imbalances to be identified.

Unit Price Distribution: The unit price distribution was analyzed using a Kernel Density Estimate (KDE) plot. Separate distributions were generated for sales and returns to compare transaction behaviors across price bands. This approach allowed for a smooth visualization of density peaks and tails, highlighting concentration zones and divergence patterns between the two transaction types.

Sales Trend Calendar: Sales Trend Calendar was created by aggregating data at the quarterly and monthly level and organized into a rows-by-columns matrix to preserve temporal structure. Quarterly aggregates are plotted as grouped bars to compare year-over-year performance. To understand the contributors for the quarterly sales, monthly totals were visualized with a heatmap to highlight intensity patterns across years. Data cleaning and explicit date parsing were applied first, and totals were calculated by year and month to ensure the table matched pivot-style numbers.

Discount Distribution: A histogram of effective discount percentages was plotted using raw sales counts on the primary axis, and a Kernel Density Estimate (KDE) curve was overlaid on a secondary axis to capture the underlying probability distribution. This dual-axis approach was chosen so that transactional volume and density could be separated for diagnostic clarity, while a combined legend was constructed to ensure interpretability. Transactions with discounts greater than 99% and sale prices below ₹10 were filtered and flagged so that potential free giveaways or zero-revenue items could be isolated. This enabled a layered view of discount behavior, balancing volume analysis with pricing sensitivity and margin exposure.

SKU Velocity: SKU velocity was calculated as average units sold per day, and realized margin was computed from transaction-level net revenue minus cost. Each SKU was classified into movement tiers based on velocity thresholds (deadstock (< 0.01 units/per day, slow-moving (between 0.01 and 0.3 units/per day, fast-moving (> 0.3 units/per day), and plotted on a scatter chart with velocity on the x-axis and margin on the y-axis. Category coloring was applied to make segment boundaries visually distinct and to surface outliers that deviate from typical performance patterns.

Sales Basket Modeling: During the initial stage of the project, it was observed that products were recorded under different names and spellings, which led to inconsistencies in the dataset. An AI based automatic mapping like *Fuzzy string matching*, *Vector embeddings* and *Clustering / nearest-neighbor search* were then attempted so that similar product names could be aligned and reduce duplication. Although this approach provided a useful first pass, certain mismatches remained due to spelling variations and domain-specific nuances. To ensure accuracy, a manual mapping exercise was subsequently carried out, through which products were standardized with correct spellings and consolidated categories using product mapping lookup table.

In order to focus the analysis on high-impact items, a threshold was applied whereby only products with sales greater than 100 units were retained. Through this filtering and mapping 96 products to 31 corrected names, approximately 85,000 of the 93,000 records were categorized and mapped correctly, representing the majority of the dataset. Once the data had been cleaned and standardized, a market basket analysis using *Apriori algorithm* was conducted. Confidence, lift, and support metrics were calculated to identify strong product affinities. Finally, the results were visualized through network graphs. The *Kamada-Kawai layout* was employed to achieve balanced spacing, while edge color was used to represent *lift*, edge width was used to represent *confidence*, and node shape was used to distinguish *single products* from *product sets*.

Net Profit Analysis: Monthly realized profit was computed by merging sales transactions with SKU-level unit cost, multiplying unit cost by sold quantity to derive COGS, and subtracting COGS from net sales after discounts. Returns were handled by allowing negative quantities to reverse COGS

and net sales appropriately. Results were aggregated by month to produce realized profit, and operating expenses were merged on month to derive net profit. A stacked area chart was plotted with realized profit above zero, expenses below zero, and a net profit line overlay to visualize the margin-expense relationship over time.

***Operational Insight*:** Sales and inventory records were examined to identify how items are tracked from receipt to sale. It was observed that manufacturer barcodes are identical across batches, which prevents differentiation of stock received at different times. Without batch identifiers, the system cannot automatically enforce FIFO logic.

An evaluation was carried out on four possible approaches to introduce batch-level tracking alongside manufacturer barcodes. The options considered were: (1) adding an additional batch barcode next to the manufacturer barcode, (2) creating a complete customized barcode with embedded batch information, (3) using multi-barcode scanners to read both manufacturer and batch codes at the same time, still requires additional batch barcode but does not require specific placement (4) implementing RFID tags that carry batch identifiers. Each option was assessed for feasibility, cost, and operational impact, with attention to alignment with existing POS system.

2.3. Tools

This project is built entirely in ***Python***, selected for its flexibility, rich ecosystem, and precise control over data workflows. The program follows a plug-and-play architecture, anchored by a centralized singleton app container. This container manages configuration, a pre-processed and enhanced dataset cache, and a growing collection of analyzer tools. Reusable services are coded to handle input file reading, csv/xlsx file exports, chart abstraction, and other utilities, such that new analyzers can be integrated seamlessly as they are developed. With the goal of keeping the tool light weight yet fully functional, Python command line interface appeared best choice. Figure **FSF-02** is a sample output of the Menu driven Python Command Line Application developed for the Data Analysis Engine.

```
(.venv) PS B:\Source\Python\BDM\Eesbee\project> python .\source\main.py
Preparing datasets...

Welcome to the Data Analysis Engine. Please select the Analyzer type you are interested in...

Available options:
1. STATISTICS
2. BASKET
3. MARGIN
4. PROFIT
5. SALES
6. INVENTORY
7. TEST_BED

Type the analyzer number of your choice, or enter 'exit' to quit.
>> 1
Selected Analyzer type: STATISTICS
Computing Statistics...
✅ Analysis complete. Results saved to the output folder.
>> 3
Selected Analyzer type: MARGIN
Performing Margin Analysis...
✅ Analysis complete. Results saved to the output folder.
>>
```

Figure **FSF-02**: Menu driven Python Command Line Application for the Data Analysis Engine

The application is modular, scalable, and offers full control over formatting and output structure, essential for business-facing diagnostics. **R** and **Google Colab** were initially considered but ruled out due to limitations in flexibility, cloud dependency, and lack of packaging support for long-term business deployment. Figure **FSF-03** is the high-level system architecture diagram of the developed Python Framework.

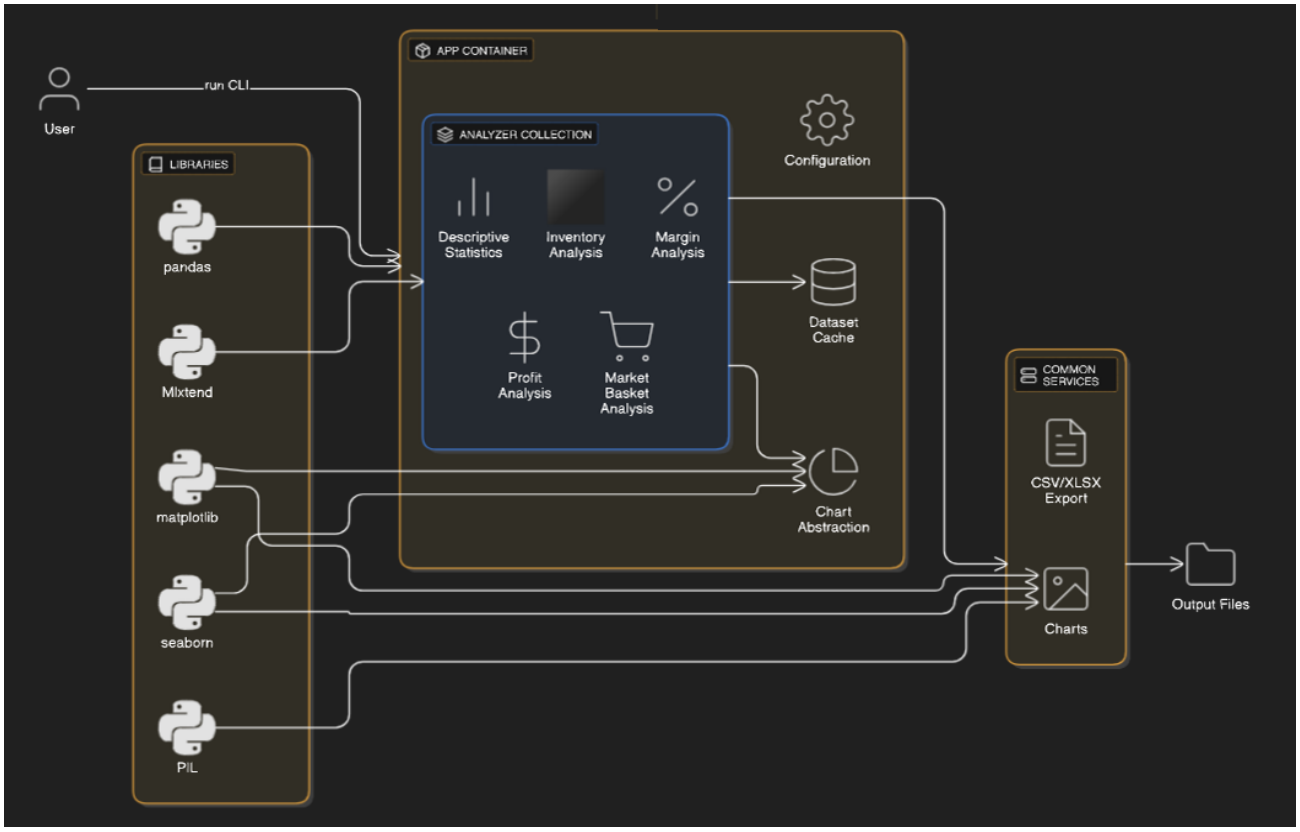


Figure **FSF-03**: High level System Architecture Diagram for Python Command Line Application

The entire code based is uploaded to the BDM Project Sep 2025 folder on the google drive and is available at the following link:
https://drive.google.com/drive/folders/1wg_f42GfTeMN3rEyTI3y-qcsgQIa793S?usp=sharing

3. Results and Findings

Full result summary tables, complete computed raw datasets and generated plots are stored in the corresponding analysis files.

3.1. Margin Diagnostics

Computed margin raw datasets and generated plots are stored in the *result_margin.xlsx* file. Various diagnosis outcomes are discussed below.

Margin Statistics: Margin statistics for *Realized Margin, Unrealized Marign, Margin Gain and Margin Erosion* are summarized in Figure **FSF-04**. The statistics are computed for the period of 30 months, starting April 1st 2023 till September 30th 2025. While majority of the metrics look healthy,

large capital is tied to the inventory and is reflecting in more than ₹40 lakhs of unrealized gain. This tied up capital needs to freed and further sections discuss strategies to address this concern. All these metrics should be monitored regularly, unrealized value prioritized for conversion, and erosion drivers investigated to protect overall profitability.

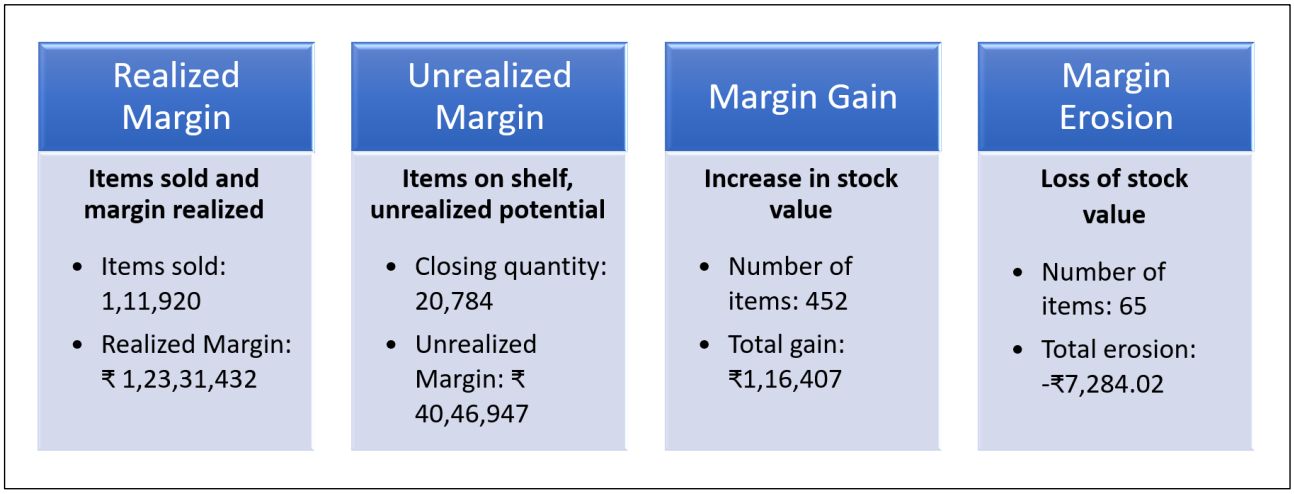


Figure **FSF-04**: Details of Margin Statistics

***Segmented Contribution & Margin Efficiency*:** Segmented Contribution of Sales and Realized margin allowing with Margin Efficiency overlay is shown in Figure **FSF-05**.

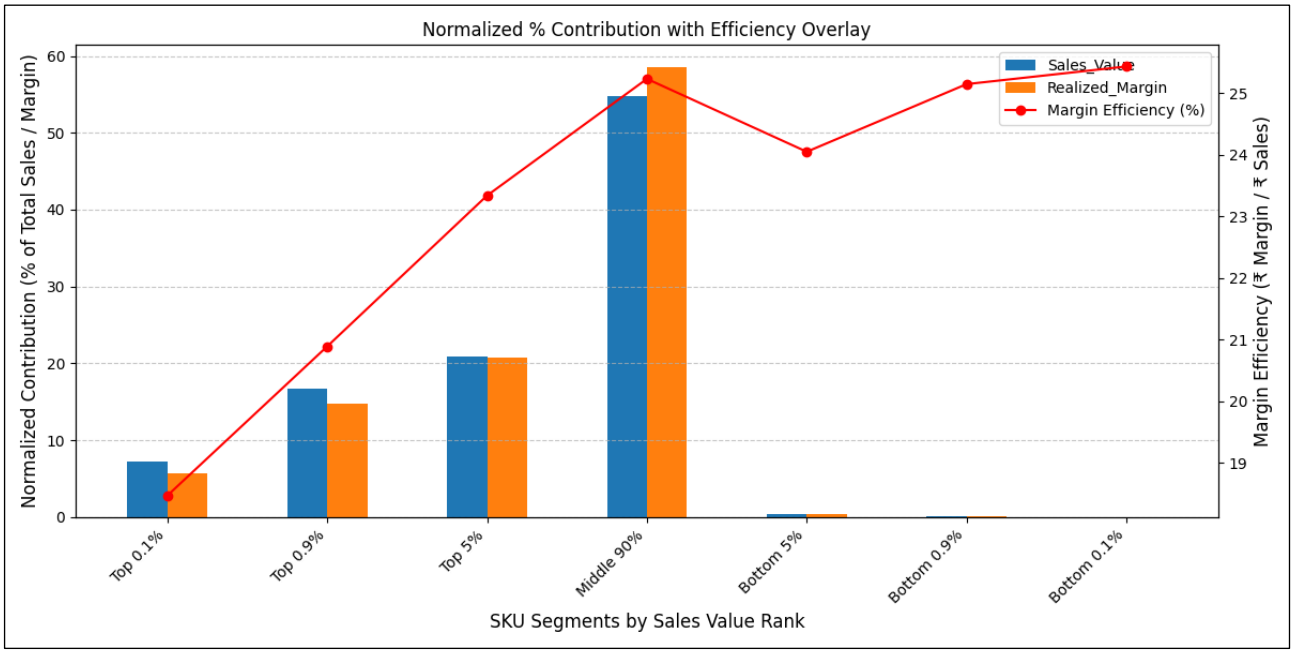


Figure **FSF-05**: Segmented Contribution & Margin Efficiency

The top-performing SKUs performed well and contributed to the overall sales and margin. However, the middle 90% of SKUs were found to drive most of the sales and margin, confirming that this group forms the core of the assortment. In addition, this group has high margin efficiency, a reason for business to focus in this area. In the bottom segments, efficiency appeared high, but this was mainly due to very low sales, even a small margin can look impressive when sales are minimal.

Top performers: Figure **FSF-06** shows Top 5 for Realized Margin, Sales Quantity and Sales Amount Leaderboards.

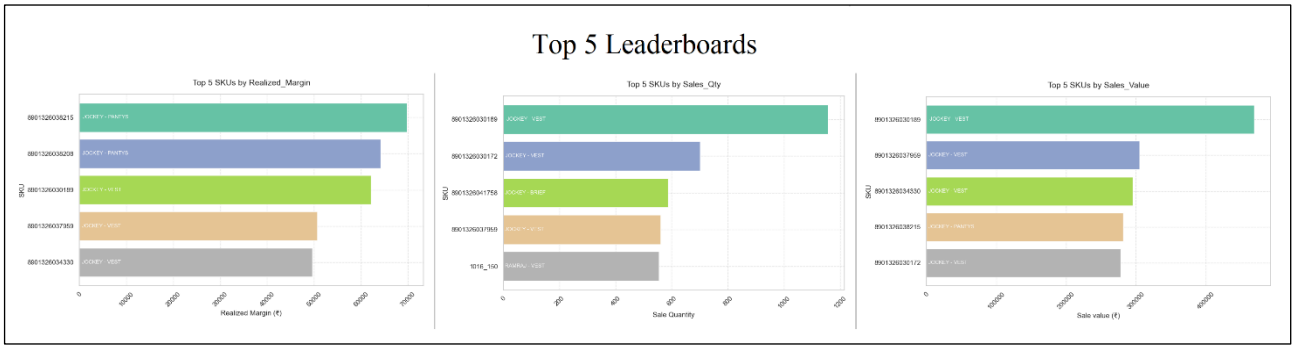
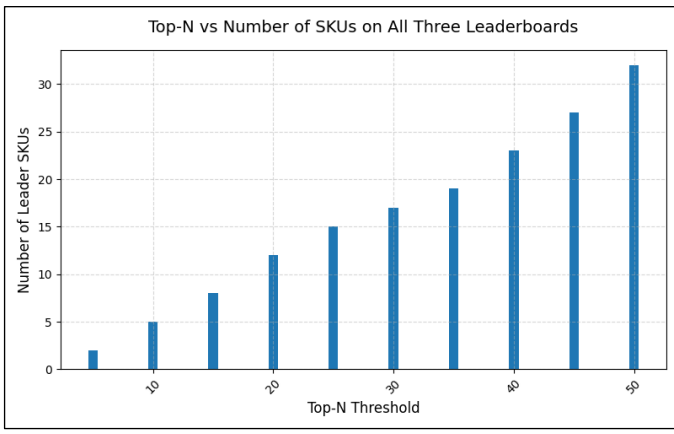


Figure **FSF-06**: Top 5 Realized Margin, Sales Quantity and Sales Amount Leaderboards

Looking across margin, sale quantity and value, a few SKUs really stand out. The top seller by both quantity and value is 8901326030189 (JOCKEY - VEST), but interestingly, it doesn't show up in the top margin list, which probably means it's priced too low. On the flip side, 8901326038125 (JOCKEY - PANTYS) has the highest realized margin but barely sells, so there's likely untapped potential there. Then there's 8901326037959 (JOCKEY - VEST), which performs well across all three metrics, a solid all-rounder worth protecting and scaling. Overall, there's a clear opportunity to rethink how Eesbee can promote and price these SKUs: push the high-margin ones harder, dig into cost structures for the high-volume ones, and double down on the balanced performers.

Top performers - Enhanced: The above Top 5 analysis identified just two SKUs that appear on all three leaderboards. This offers limited insight and often misses high-potential SKUs that emerge just outside the top ranks. By expanding the threshold incrementally up to Top 50, a more comprehensive review process that captures both dominant performers and rising contenders can be facilitated. This approach helps the business identify which SKUs are persistently strong across all leaderboards, which ones show late-stage potential, and where promotional or pricing strategies might unlock untapped value. Figure **FSF-07** illustrates how broader thresholds surface more leader SKUs, reinforcing the need for a scalable diagnostic lens. Full results are saved in *result_margin.xlsx* file.



Sample SKU Leaders and their Top Buckets

SKU	Brand	Product	Top_N Buckets
8901326030172	JOCKEY	VEST	[15, 20, 25, 30, 35, 40, 45, 50]
8901326030189	JOCKEY	VEST	[5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
8901326030264	JOCKEY	VEST	[45, 50]
8901326030271	JOCKEY	VEST	[15, 20, 25, 30, 35, 40, 45, 50]
8901326030288	JOCKEY	VEST	[45, 50]
8901326030868	JOCKEY	PANTY	[20, 25, 30, 35, 40, 45, 50]
8901326030875	JOCKEY	PANTYS	[20, 25, 30, 35, 40, 45, 50]
8901326030882	JOCKEY	PANTYS	[40, 45, 50]
8901326033104	JOCKEY	BRIEF	[25, 30, 35, 40, 45, 50]
8901326034323	JOCKEY	VEST	[45, 50]
8901326034330	JOCKEY	VEST	[10, 15, 20, 25, 30, 35, 40, 45, 50]
8901326034446	JOCKEY	BOXER BRIEF	[50]
8901326037942	JOCKEY	VEST	[25, 30, 35, 40, 45, 50]
8901326037959	JOCKEY	VEST	[5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
8901326038031	JOCKEY	BRIEF	[45, 50]

Figure **FSF-07**: Top-N vs Number of SKUs on All Three Leaderboards

3.2. Inventory Analysis

Computed raw datasets for inventory analysis are stored in the *result_inventory.xlsx* file. Various diagnosis outcomes are discussed below and are shown in the Figure **FSF-08**.

Negative Stock Report: Few SKUs are showing negative closing balances across brands like VIP, RAMRAJ, VSTAR etc. These findings need to be updated in the system by the store management for accurate inventory position. The report is saved in *result_inventory.xlsx* with tab name *Negative Stock*.

Stocked but Unsold Returns Report: A total of **2,230 SKUs** was stocked but show *no sales and no closing inventory*, indicating they were likely returned after shelving. These items ***occupied shelf space and operational effort*** without contributing to revenue, with a cumulative return value of **₹14.28 lacks**. Sharing this insight with business to support inventory efficiency improvements, targeting these SKUs can help reduce non-converting stock and optimize assortment planning. The complete list is saved in *result_inventory.xlsx* with tab name *Stocked Then Removed*.

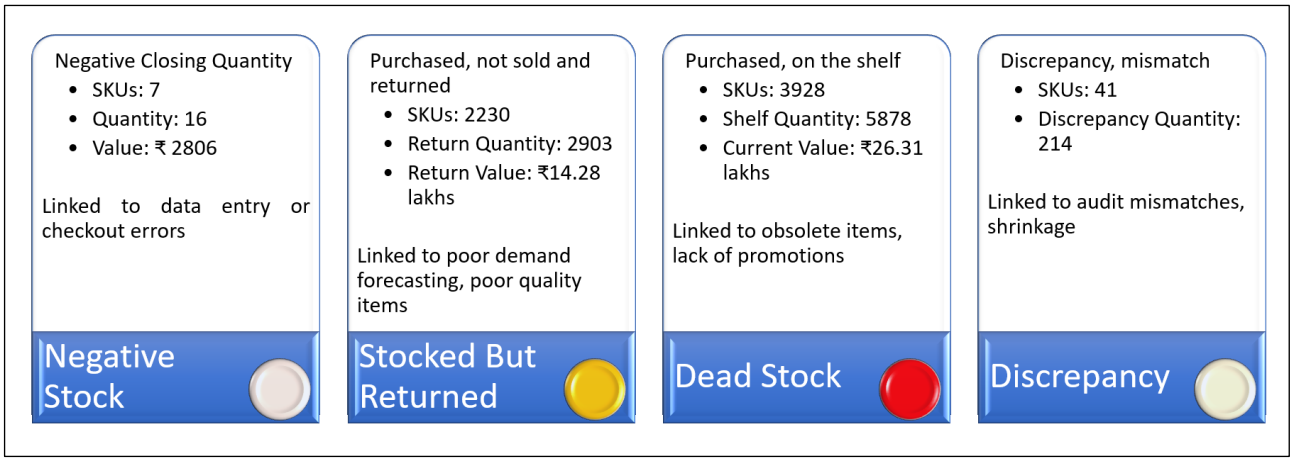


Figure **FSF-08**: Inventory Analysis and Categorization

Dead Stock Report: The deadstock report found **3,928 SKUs** with a total purchase value of **₹26.31 lakhs** and unrealized margin of **₹14.47 lakhs**, reflecting significant capital tied up in non-moving inventory. Many of these items show recent purchases but zero sales, indicating poor sell-through and potential overstocking. These SKUs should be prioritized for markdowns, liquidation, or repurposing to recover value and improve inventory efficiency. The complete list is saved in *result_inventory.xlsx* with tab name *Dead Stock*.

Discrepancy Stock Report: The discrepancy stock report identified **41 SKUs** where actual closing quantities deviate from expected values, despite clear purchase and sales activity. These gaps may stem from mis-scans, unrecorded returns, or verification errors, and should be prioritized for physical audit and system correction to maintain inventory integrity. The complete list is saved in *result_inventory.xlsx* with tab name *Discrepancy Stock*.

Inventory Flow Diagnostics: The *Stock Reconciliation Waterfall* graph shown in Figure **FSF-09** reveals that purchases and sales volumes were broadly balanced over the 30-month period, yet inventory accumulated modestly from 13,984 to 20,784 units.

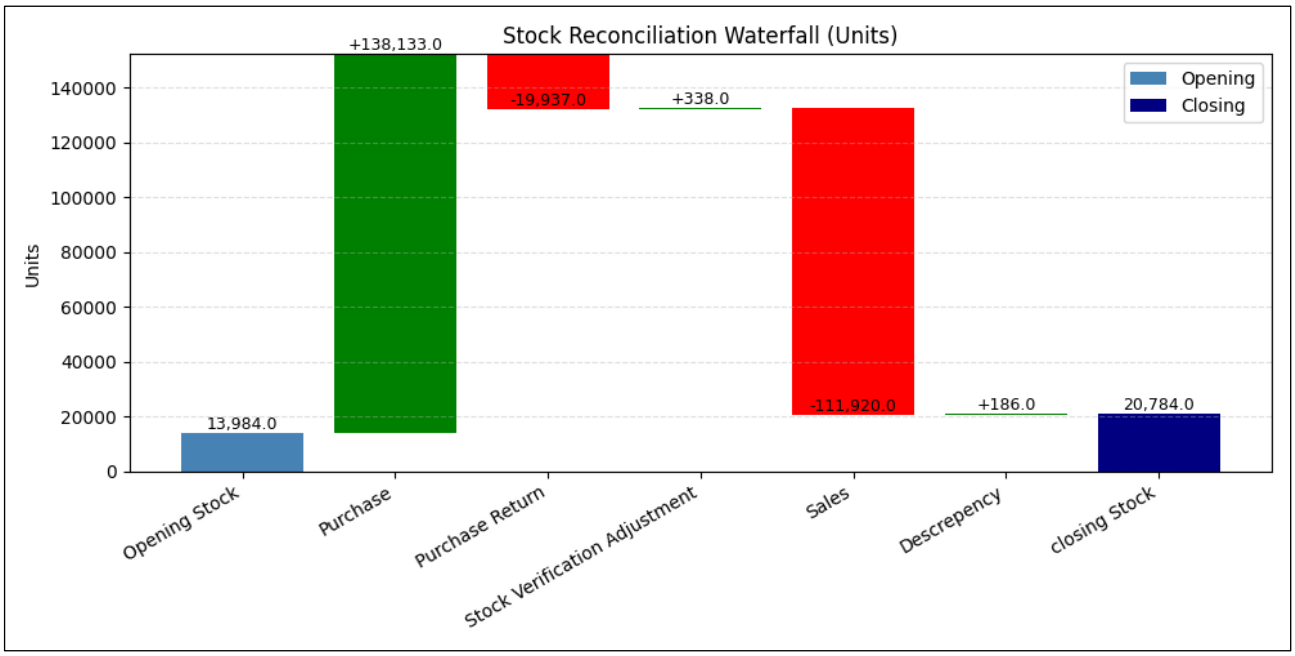


Figure **FSF-08**: Stock Reconciliation Waterfall

To evaluate whether this pattern is healthy, key ratios were computed using unit-based formulas and compared against published benchmarks for Indian retail. Generally, the ratios use the amounts, like Cost of Goods Sold. However, in the case just the unit metrics are used as they seem appropriate for the inventory analysis. The details and corresponding insights are document in the Table **FST-01** given below. The analysis reveals that while procurement aligns with sales in scale, the pace of inventory turnover is significantly below industry standards. This indicates that stock is being held longer than expected, tying up capital and increasing the risk of aging inventory.

Name of Metric	Formula	Computed Value	Industry Standard	Insight	Reference Link
Inventory Turnover (units)	Units Sold ÷ Average Inventory	111,920 ÷ 17,384 ≈ **6.4 turns in 2.5 yrs (~2.6 per yr)**	6–10 per year (Fashion/General)	Turnover is **below benchmark**, showing slow-moving stock relative to demand.	Shopify India Retail Guide
Purchase-to-Sales Ratio	Net Purchases ÷ Sales	118,196 ÷ 111,920 ≈ **1.05**	1.0–1.1 (healthy alignment)	Procurement volume is aligned with sales; issue lies in timing/velocity, not overbuying.	Megaventory Retail Benchmark
Inventory-to-Sales Ratio	Average Inventory ÷ Annual Sales	17,384 ÷ 44,768 ≈ **0.39**	~0.65 (Corporate India FY24 stock-to-sales ratio)	Ratio is leaner than benchmark, but combined with low turnover suggests inventory is not cycling fast enough.	Financial Express – Corporate India Stock-to-Sales

Table **FST-01**: Inventory Management KPIs and comparison to industry averages

3.3. Sales Analysis

Computed raw datasets for inventory analysis are stored in the *result_sales.xlsx* file. Various diagnosis outcomes are discussed below.

Unit Price Distribution: The unit price distribution KDE plot shown in figure **FSF-09** revealed that sales were concentrated in lower price bands, with a smooth density peak indicating consistent movement of affordable items. In contrast, the return distribution exhibited a longer tail and higher density in premium price segments, suggesting that higher-priced products were more prone to being returned. This divergence points to potential mismatches in customer expectations, product quality, or perceived value.

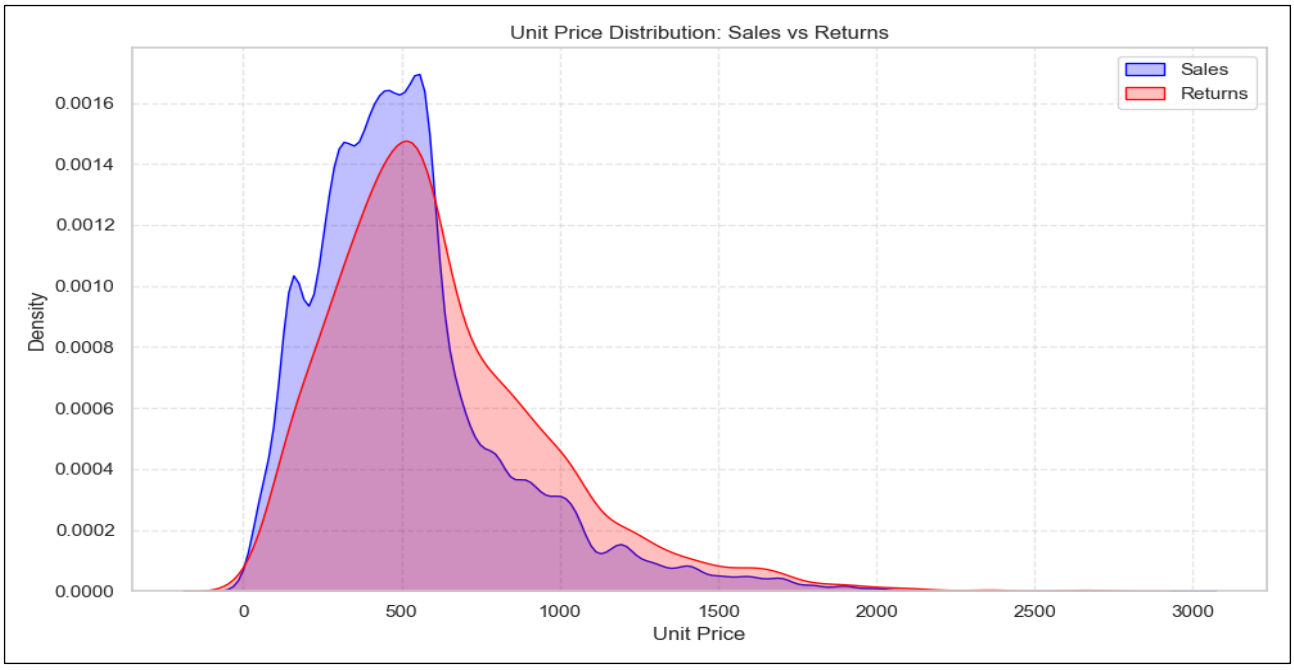


Figure **FSF-09**: Unit Price Distribution

Sales Trend Calendar: The Quarterly Sales Trend Calendar in figure **FSF-10** shows a consist year-over-year improvement across all quarters for the time period April 1st 2023 to September 30th 2025.

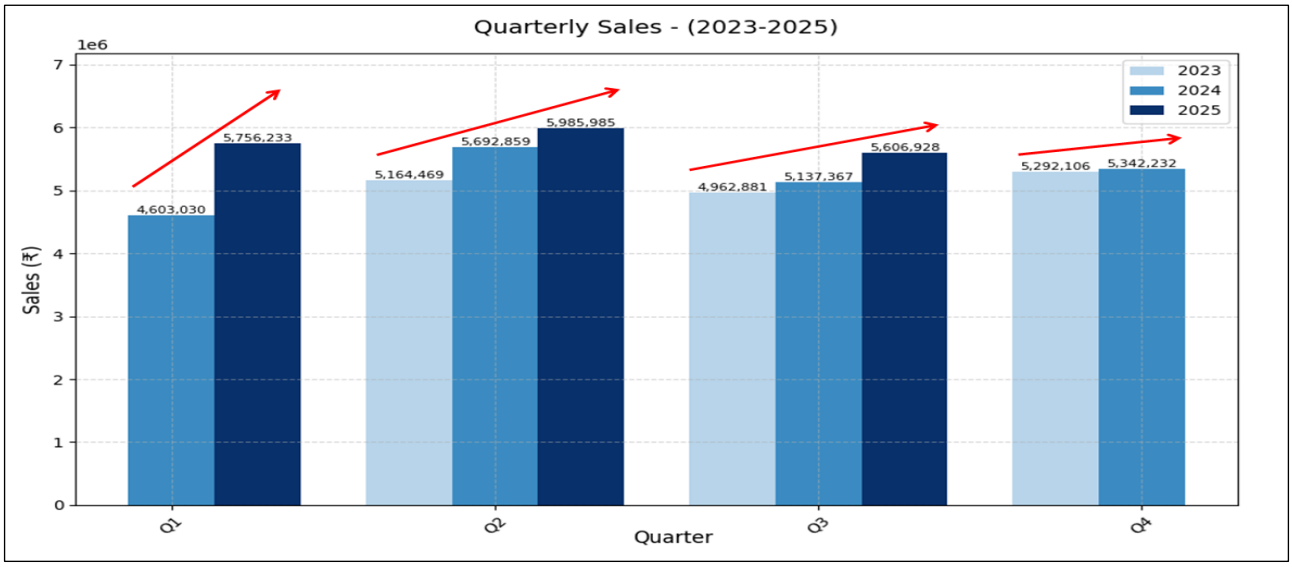


Figure **FSF-10**: Quarterly Sales Trend Calendar

However, seasonal peaks were observed in specific months and were found to drive quarter totals unevenly. To establish this unevenness, the Monthly Sales Trend Calendar is plotted as a heatmap and is shown in Figure **FSF-11**.

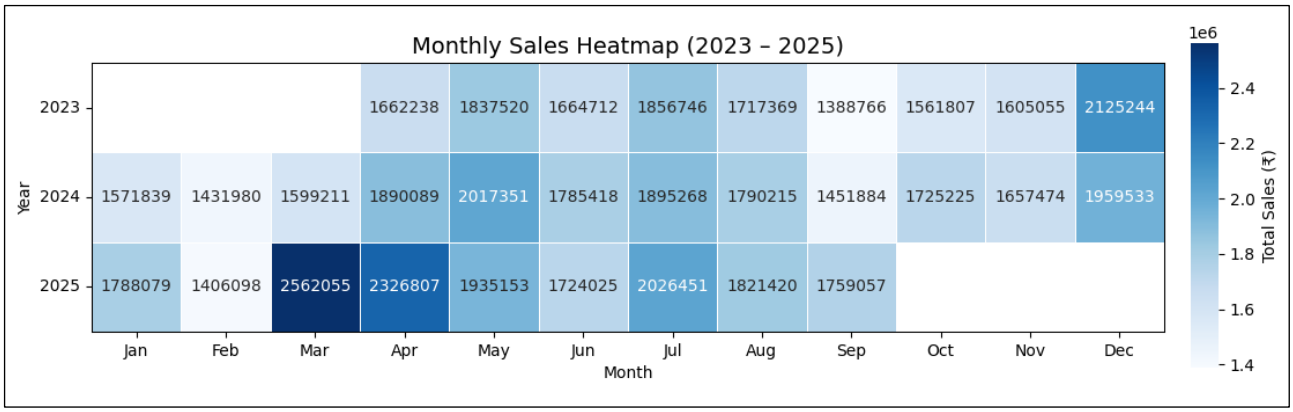


Figure **FSF-11**: Monthly Sales Trend Calendar

March 2025 was observed as a clear outlier with significantly higher sales, while March 2024 was weak. The successful tactics used for March 2025 should be documented and repeated, with inventory and promotional plans locked earlier. Similarly, April 2025 fares much better when compared to April 2024. September was low in 2023 and 2024 but showed improvement in 2025, so it should be standardized as a repeatable campaign. Recurrent softness was seen in June and August across the period, so should be supported by focused, short promotions and by ensuring full size/color availability for top SKUs. Quarterly comparisons revealed that while overall totals increased year over year, the composition of each quarter varied — with some driven by sustained multi-month contributions and others by isolated monthly surges.

Discount Distribution: Distribution of the discount percentage is plotted in the Figure FSF-12.

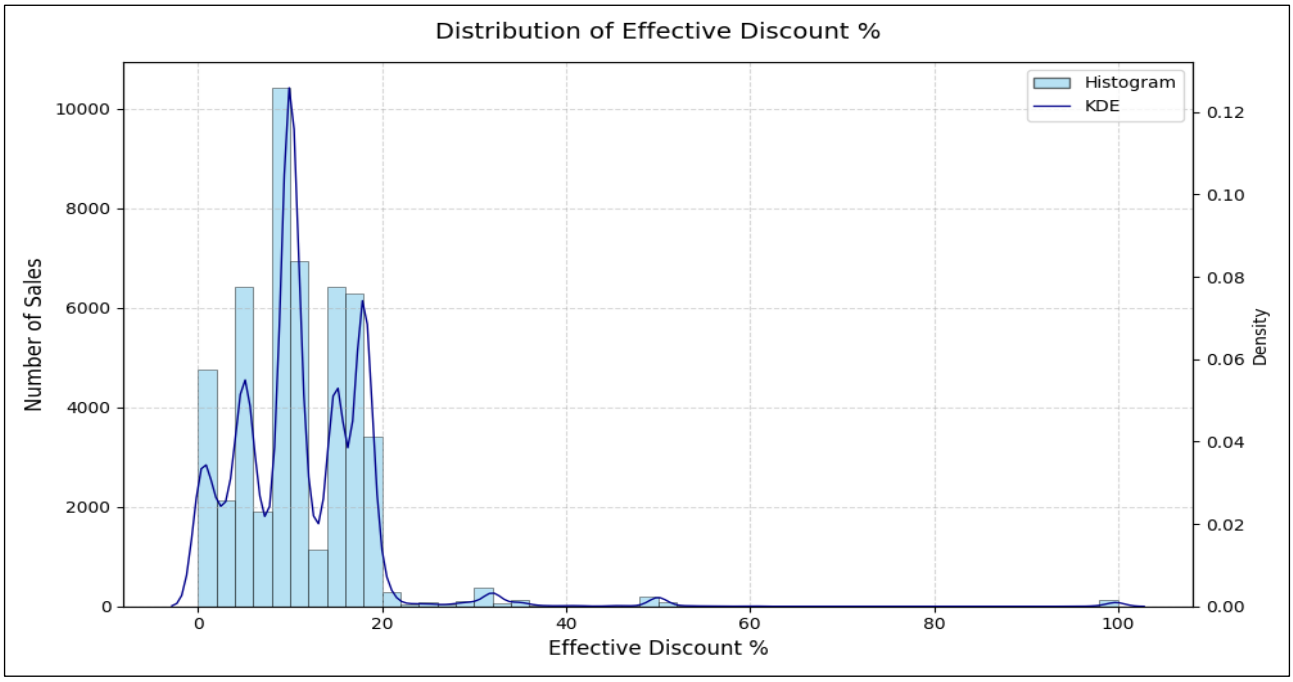


Figure **FSF-12**: Distribution of Effective Discount Percentage

A concentrated band of sales activity was observed between 5% and 15% discount, where both histogram volume and KDE density reached their peak, marking this range as the most commercially effective. Beyond 20%, both metrics were seen to decline sharply, suggesting reduced customer responsiveness and margin erosion. A small but distinct spike was detected at the 100% discount level, corresponding to near-zero sale prices, which highlighted the presence of free giveaways or operational anomalies. Although these transactions were low in volume, they represented complete margin loss and required separate tracking. *128* sale records with retail value of *₹1.15 lacks* were flagged based on discount > 99 % or Sale value < ₹10 for business review. Overall, clear behavioral patterns in discount utilization were surfaced, along with outlier segments that could distort pricing diagnostics.

SKU Velocity: SKU Velocity was categorized as fast moving, slow moving, and deadstock and is plotted against Realized Margin as shown in the Figure **FSF-13**.

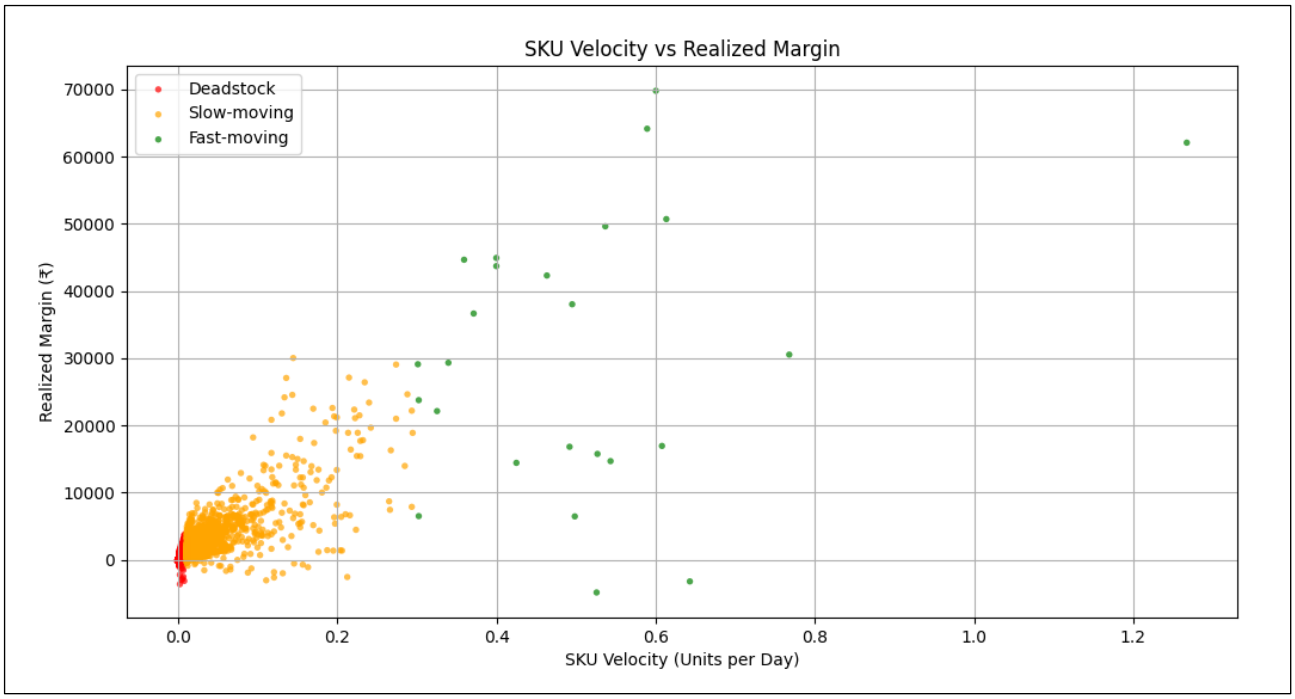


Figure **FSF-13**: SKU Velocity vs. Realized Margin

A dense cluster was observed near the origin, indicating many SKUs with low velocity and low realized margin, consistent with deadstock or slow-moving profiles. Fast-moving SKUs were found to be more dispersed across the plot, with a tendency toward higher margins and higher daily units, suggesting that volume and profitability often reinforced each other. The separation of tiers appeared meaningful, with a small set of fast-moving items driving disproportionate value while a long tail of low performers accumulated holding and opportunity costs.

Sales Basket Modeling: Sales Basket modeling using *Apriori algorithm* resulted in many basket item matchings. To keep it clean, Figure **FSF-14** is plotted for one-to-one network for single-item

relationships clustered around a few staple SKUs, and Figure **FSF-15** is plotted for many-to-one network for multi-item relationships.

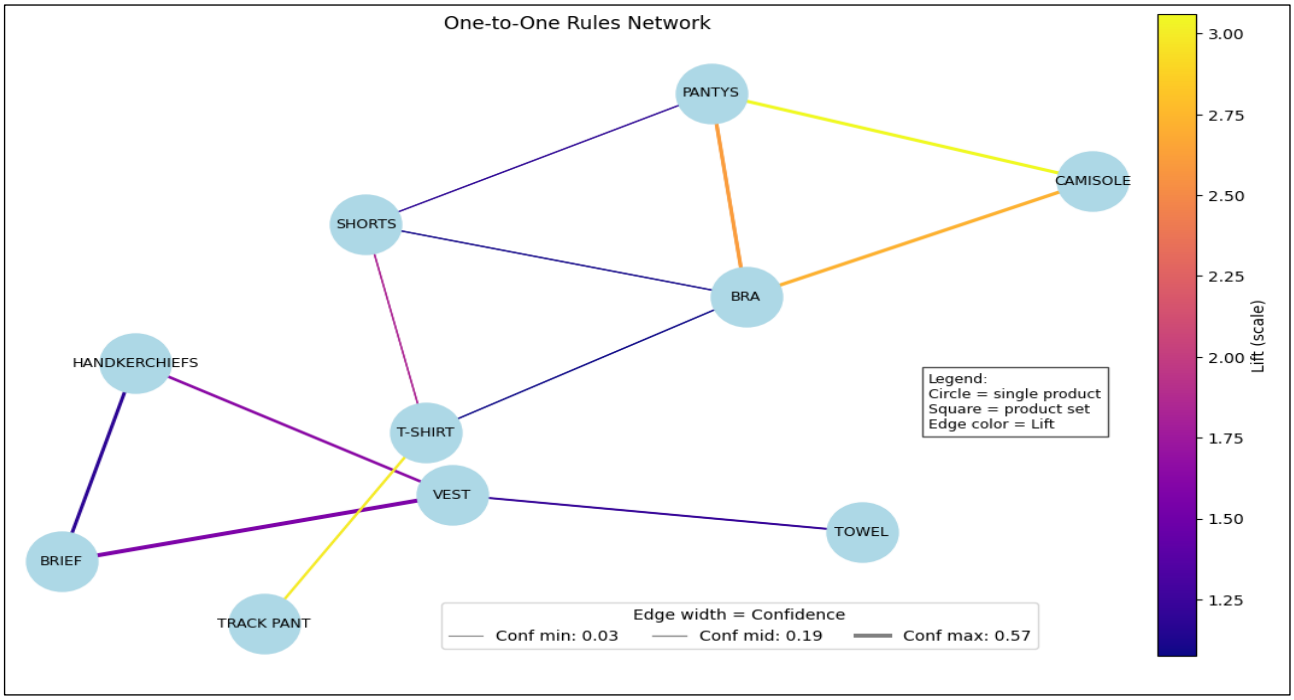


Figure **FSF-14**: Market Basket Analysis: One-to-One Rules Network

Based on one-to-one network, it can be observed that *pantys* frequently appearing as a central node linked to *briefs*, *bras*, and *T-shirts*. These pairwise connections were characterized by moderate lift and varied confidence, indicating routine, small-basket behavior rather than rare or highly specialized pairings. Because the links were consistent across many simple transactions, they were interpreted as reliable signals for straightforward cross-sell opportunities.

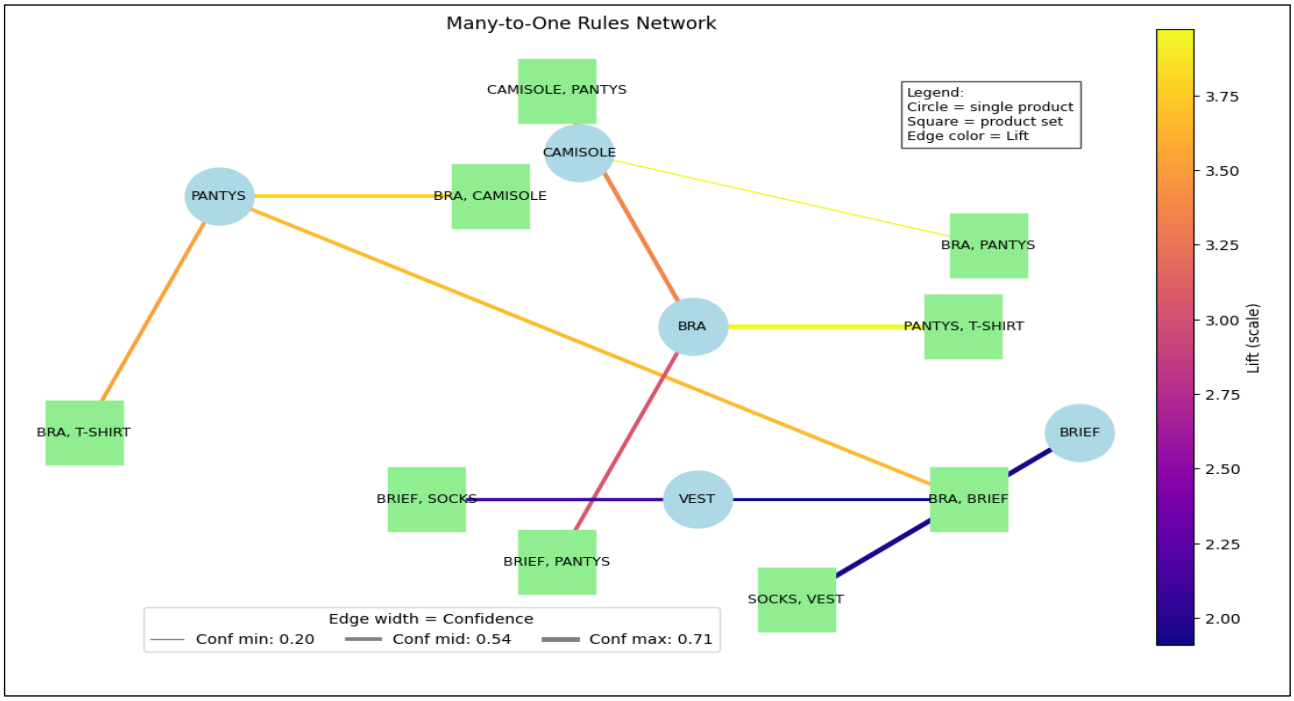


Figure **FSF-15**: Market Basket Analysis: Many-to-One Rules Network

In the many-to-one network, multiple antecedent combinations were found to converge on the same destination SKUs, most notably bras, where lift and confidence tended to be stronger. Combinations such as camisole + pantys → bra and socks + vest → bra were repeatedly present, suggesting that more deliberate, multi-item shopping missions often culminated in a single target product. This pattern was taken to reflect both replenishment behavior and complementary purchase flows that could be leveraged with contextual prompts.

Taken together, the two visualizations were interpreted as showing a dual shopper dynamic: frequent, simple purchases anchored by high-frequency basics, and slightly larger, goal-oriented baskets that end at destination items. The strength and consistency of these signals were judged sufficient to support targeted, small-scale tests, while it was recommended that rules be validated across time and segments before broader operational rollout.

3.4. Profit Analysis

Figure **FSF-16** shows the plot of Net Profit over 30-month period, from April 2023 to September 2025. A pronounced mid-year peak in net profit was observed alongside strong realized margins, while a late-year trough coincided with margin compression and relatively stable expenses.

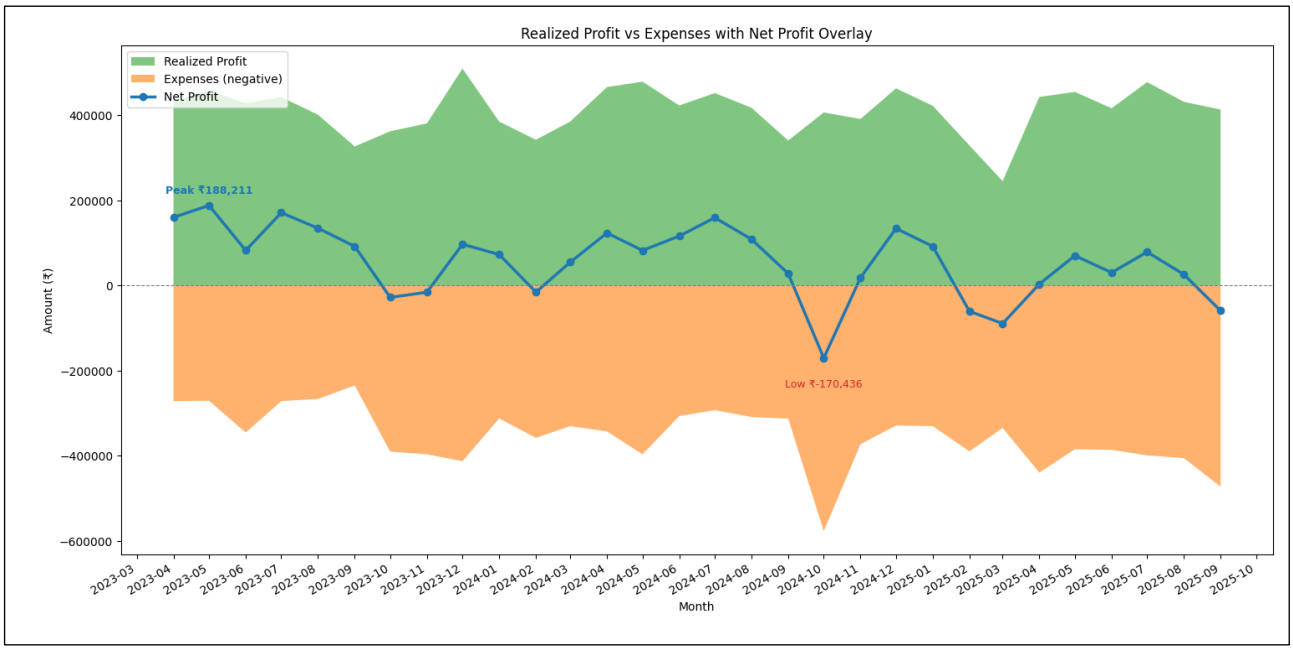


Figure **FSF-16**: Realized Profit vs. Expenses with Net Profit Overlay

The spread between realized profit and expenses narrowed into the trough, indicating that profitability was predominantly constrained by margin rather than by cost escalation. A partial recovery into early 2024 was noted, but peak levels were not fully regained, suggesting lingering effects from discounting, returns, or lower-margin mix. October 2024 recorded the lowest figure, primarily due to the unusually high expense that month.

3.5. Operational Insights

Batch tracking is an important and pressing operational problem to the Eesbee. Multiple solutions were identified, but it was observed that each solution carries trade-offs. The current POS system scans one barcode per item at the checkout, hence complete information should be embedded into that one scan.

The augmented batch barcode in conjunction to the manufacturer barcode requires precise placement and alignment, which complicates the packaging. The solution to create a customized barcode invalidates the manufacturer's code and create compliance risk. Multi-barcode scanners provide flexibility but require POS and software upgrades. RFID offers automation and scalability, but is capital-intensive. The following Table **FST-02** compares different solutions and their limitations.

Criteria	Augmented Barcode (Extra Label)	New Customized Barcode	Multi-Barcode Scanning	RFID Tags
Implementation Cost	Low (printing labels)	Medium (system + label redesign)	Medium (scanner + POS upgrade)	High (hardware + tags)
Ease of Adoption	Moderate (placement issues)	Low (invalidates manufacturer code)	Moderate (training + system update)	Low (requires infrastructure)
POS Compatibility	Works with current scanners	Works with current scanners	Requires POS upgrade	Requires RFID readers
Traceability	Good (batch ID added)	Good (embedded batch info)	Good (dual scan)	Excellent (automated batch tracking)
Scalability	Limited (manual labeling)	Poor (custom codes per item and batch)	High (once POS updated)	Very high (bulk scanning, automation)
Risks/Issues	Label alignment, human error	Compliance risk, cumbersome	Training, system upgrade	High upfront cost

Table **FST-02**: Batch Identification Solutions Comparison

This analysis highlights that multiple technical solutions exist, but the choice must balance cost, compliance, and ease of integration with the current systems. Given the rigidness of the POS system and the setup for single barcode scan, Eesbee does not have an out-of-box cost effective and scalable solution at the present time. Depending on the long-term objective, Eesbee can chose one of the above solutions, or a hybrid solution that can mix-and match per the requirement.

4. Interpretation of Results and Recommendations

Based on the analysis and insights gained, following are few recommendations to the Eesbee management for ***Optimizing of Sales, Inventory and Margin*** and to improve overall health of the business. The recommendations and action items are categorized into *Operational*, *Technical* and *Marketing* buckets, and are documented in the below **Table FST-02**.

	Sl. No.	Insights	Recommendations	Action Items
Operations	1	Margin Statistics	A focused set of actions is recommended to protect realized margin, convert unrealized value, scale margin gains, and stop margin erosion.	• Replenishment priority and safety stock levels should be raised for the top SKUs by realized margin • A targeted promotion and merchandising plan should be executed for high-value, low-velocity SKUs
	2	Top Performers and Top Performers-Enhanced	Reassess pricing strategies for high-volume SKUs to improve margin contribution. In addition, continue to protect Balanced Performers for multiple metrics to scale further, ensuring they continue to anchor both revenue and margin resilience.	• High-margin SKUs with low sales should be promoted more aggressively to unlock potential profitability. • for balanced performers, work with suppliers for bulk ordering and volume discounts. • Review late surge SKUs for potential elevation to top performers
	3	Segmented Contribution & Margin Efficiency	Assortment and pricing strategies should focus on the middle 90% of SKUs, which were shown to drive the majority of both sales and margin.	• Review the middle 90% of SKUs for pricing, promotion, and inventory depth opportunities • Identify bottom-tier SKUs and consider bundling, repositioning, or phase-out based on strategic fit
	4	Inventory Flow Diagnostics	Inventory turnover should be raised from ~2.6 per year to at least **4–6 per year**, which is the healthy benchmark for Indian fashion retail. By explicitly setting a turnover target, management can ensure that stock cycles faster, capital is freed up, and aging risk is reduced.	• Focus on targeted promotions, markdowns, and faster replenishment cycles for slow-moving SKUs. • Purchase orders should be paced closer to actual demand, with smaller, more frequent replenishments instead of bulk buys.
	5	Unit Price Distribution	Review pricing strategy and product positioning to ensure	• Capture reasons for higher price item returns

			alignment with customer expectations.	• Review and Refine return policy for premium segments, to mitigate losses and improve customer satisfaction.
	6	Sales Trend Calendar	March and April 2025 produced record sales and the execution plan should be repeated. June and August seems seasonally low, so planning for slow months is essential.	• Design **targeted campaigns** for historically weak months and reinforce supply chain readiness for historically strong months. • Inventory, staffing, and marketing calendars should be adjusted to anticipate strong months and mitigate weak ones.
	7	Discount Distribution	Promotional strategies should be concentrated within the 5%–15% discount range, where customer responsiveness and sales volume were found to be highest. Deeper discounts beyond 30% should be reviewed for margin impact and commercial viability.	• Establish a clear discounting policy (could be based on sale value, customer loyalty, liquidation stock etc.) • Review ₹1.15 lacks of free giveaways to make sure they are valid and not a technical or operational issue
	8	Negative Stock	Immediate reconciliation of negative stock should be prioritized to prevent fulfillment errors and financial misstatements.	• Run a daily negative-stock report and reconcile discrepancies with POS transactions.
	9	Unsold Purchase Returns	Returned inventory should be triaged and processed to recover value and reduce holding costs.	• Implement a returns disposition workflow that routes items to restock, refurbish, or liquidate within 7 days.
	10	Dead Stock	Dead stock should be cleared through targeted promotions or liquidation to free up working capital.	• Launch a time-boxed markdown and bundle campaign for identified dead SKUs and monitor sell-through weekly.
	11	Stock Discrepancy	Inventory discrepancies should be investigated and root causes addressed to improve accuracy.	• Train personal if needed to improve data entry accuracy • Spot check, cross validate frequently to minimize discrepancies
	12	SKU Velocity	Inventory and pricing actions be focused on the extremes, fast-moving SKUs and slow moving and near deadstock	• fast-moving SKUs should be prioritized for sustained availability and selective margin protection

				• Deadstock and slow-moving SKUs should be addressed through targeted markdowns, bundling, or rationalization
	13	Profit Analysis	Margin drivers should be stabilized by tightening discount controls and shifting mix toward higher-margin SKUs during periods of elevated expense.	• Expense cadence should be aligned with months of stronger margin • Variable costs should be timed to avoid amplification of troughs. • Review late-year margin compressions for consistent profit profile.
Technology	14	Database Backup	Database plays a critical role in the functioning of the business. Based on the Eesbee past experience and industry standards, failure of the database could cost tremendously to overall profitability and sales. Highly recommend to have a clear backup and recovery plan.	• Invest in the real time replication of the database to an alternative server on the premises • Implement nightly backup of the database to a cloud storage • Perform quarterly or half-yearly recovery of the database to make sure backup is working
	15	Operational Insight	Batch tracking should be introduced through either additional batch barcodes or multi-barcode scanners, depending on packaging feasibility and POS readiness.	• Pilot batch barcode labeling on selected SKUs to test geometric placement and scanning reliability. • Assess POS system capability for multi-barcode scanning and plan incremental upgrades. • Train warehouse and retail staff on handling dual identifiers and enforcing FIFO logic. • Develop cost-benefit analysis for RFID adoption in critical categories, with phased rollout strategy.
	16	Website Development	Online presence is essential for market penetration and to sustain this fiercely competitive market.	• Implement a basic mobile ready website to promote the store (read-only) as first phase • Add cart and check-out functionality for online purchases as second step
	17	Integration of the Tally and POS system	Integrate POS system and Tally system for accurate data and single point of truth	• Work with software providers to integrate POS system and Tally system • Import aggregated sales and purchase data from the POS system into Tally for

				accurate data by eliminating manual data entry
	18	Social Media promotions	Create store awareness by targeted advertising on social media	• Identify local Whatsapp groups and facebook pages to promote Eesbee store • Request known and frequent customers to share store details in their groups/networks • Research cost effective paid advertising
	19	Management KPI dashboard	Create a Key Performance Indicators (KPI's) dashboard for daily reviews and trend analysis	• Invest in custom software of enhance the POS system for daily review • Implement time series analysis for seasonal stocking
Marketing	20	Loyalty Card	Introduce customer Loyalty Card to engage and encourage returning customers	• Introduce points system that customers can redeem after accumulating certain points • Consider tier system for additional discounts for frequent customers
	21	Advertisements	Advertise in local newspaper or color flyers that can be distributed at local events	• Keeping Market basket Analysis and Inventory Analysis, create offers and clear deadstock inventory • Publish discount coupons in the advertisements to improve customer relationship and satisfaction
	22	Market Basket Modelling	*Prioritize hub SKUs*: Focus on *pantys* and *bras, vests and briefs* for initial cross-sell and placement tests *Adjust adjacency:* Place hub and destination SKUs closer in-store Bundle slow moving item with fast moving item at discounted rate	• Pick 3 pilot bundles (Combo offers for one-to-one, Triads for many-to-one) and finalize creative/placement. • Apply modest incentives (5–10%) for pilot bundles. • Create attractive advertainments considering seasonality, like back-to-school window or pre-winter promotions
	23	SMS alerts	Keeping in touch with the customers and engaging them with store promotions	• Identify each customer by capturing the mobile number • Sign them up for SMS alerts for sale receipts • Send bi-weekly/monthly or seasonal promotional alerts

Table **FST-02**: Recommendations and Action Items